DIC08
Inverter DC Characteristics and Noise Margins

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 8 (PDF pages 31–34) to obtain inverter transfer curves, switching thresholds and noise margins, then test downstream regeneration using finite-duration disturbances.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: VTC, slope −1 and NM (31–33)	01_dc_transfer	Central derivative and interpolated crossings
Step 2: vary width ratio (32–33)	$W_p/W_n = 0.75, 3, 12$	All three curves
Step 3: noise at each ratio (33)	02_noise_cascade	Positive low-state and negative high-state disturbances
Question: add CMOS receiver (34)	Second stage for every ratio	Both stages retained

3 Circuit and design derivation

Fix $W_n = 0.6\,\mu$m and $L = 0.25\,\mu$m; $W_p = 0.45, 1.8, 7.2\,\mu$m implements the three ratios. The switching threshold is defined by $V_o = V_{DD}/2$, not an individual MOS threshold. V_{IL}, V_{IH} are the two $dV_o/dV_i = -1$ crossings. Boundary outputs $V_{OH} = V_o(V_{IL})$ and $V_{OL} = V_o(V_{IH})$ give $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$. CSV files also retain the handout's ideal-rail NM approximation to keep definitions distinct. Partial derivative tests and scattered source sizes are replaced by complete two-boundary extraction and all three ratios.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not accep-

tance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

DC sweeps 0–1 V in 0.5 mV steps. The 18 ns transient prints every 2 ps. Low-state disturbances reach 0.2/0.4/0.48/0.6 V; high-state dips reach 0.8/0.6/0.52/0.4 V. Edges are 0.1 ns and plateaus 0.9 ns. This normalized test preserves the noise experiment without applying the original 1.2/1.3 V thresholds to a 1 V supply.

5 Results and specification checks

Both decks completed. Voltages and margins below are in V and use boundary output levels.

W_p/W_n	V_M	V_{IL}	V_{IH}	NM_L	NM_H
0.75	0.4214	0.3617	0.4496	0.3237	0.5100
3	0.4825	0.4291	0.5189	0.3855	0.4376
12	0.5459	0.5031	0.5925	0.4600	0.3616

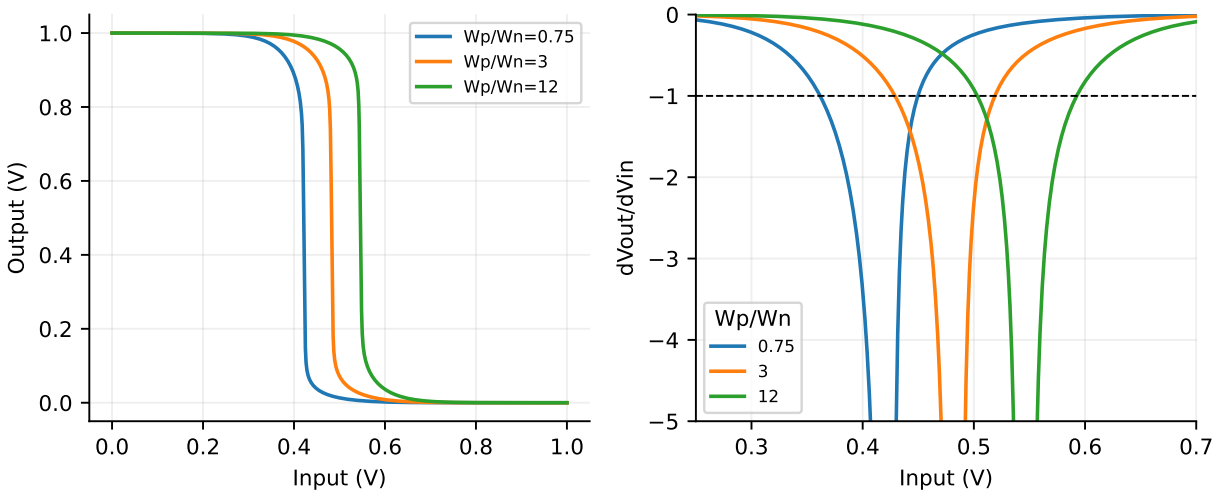


Figure 1: Transfer curves and unity-gain boundaries; redrawn from this run's HSPICE data.

6 Discussion

Widening PMOS shifts the threshold upward, increasing NM_L and decreasing NM_H; ratio 3 is not exactly symmetric in this model. A second stage can restore disturbances that stay outside its switching region, but may regenerate a wrong logic excursion into a full pulse. Filtering depends on duration and loading; transient survival is not identical to DC noise margin.

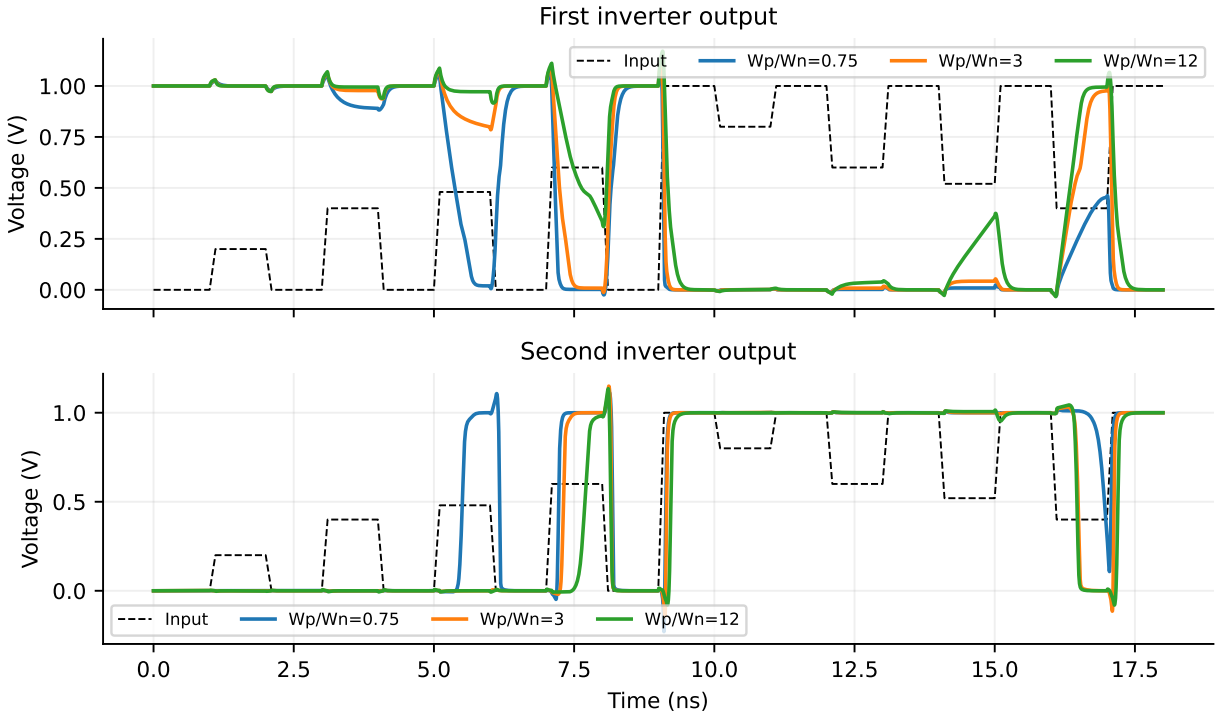


Figure 2: Noise responses at the first and second stages for all ratios; redrawn from this run's HSPICE data.

7 Conclusion

All three steps and the cascade question have complete simulation evidence. Voltage normalization and the NM definition are explicit; original-process numerical margins are not claimed.

8 MOS transistor schematics

Source: `01_dc_transfer.sp` and `02_noise_cascade.sp`. Three parallel branches use $k = 0, 1, 2$, with $W_p = 0.45, 1.8, 7.2\,\mu$m respectively; all have $W_n = 0.6\,\mu$m and $L = 0.25\,\mu$m. The DC deck contains only Xk. The transient noise deck adds XBUFk of the same dimensions to each branch.

The drawings use the device and node names of the cited netlists. Equal net labels denote an electrical connection. Unless stated otherwise, NMOS bulk terminals connect to ground and PMOS bulk terminals connect to V_{DD}. The bulk wires are omitted from the three-terminal symbols for readability.

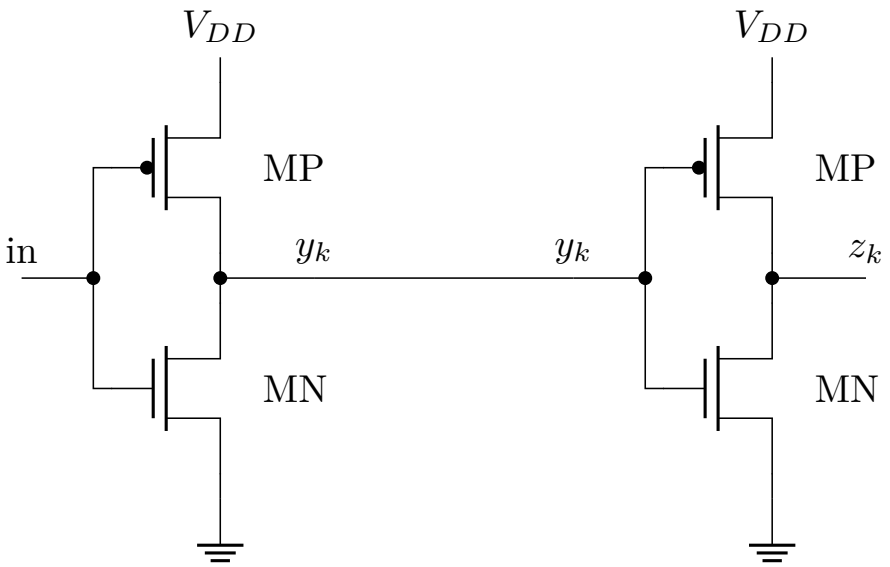


Figure 3: One complete four-MOS noise-cascade branch; repeat independently for the three width ratios. The common input does not connect the three branch outputs.